# Analysis of the acceleration of deep learning inference models on a heterogeneous architecture based on OpenVINO

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***Abstract*—Convolutional neural networks (CNN) are a very powerful tool for many different applications. This capability is highly demanded in the field of embedded systems for video surveillance, speech recognition, and image analysis. Due to its high computational intensity, the application of CNN is limited to real-time research areas where computational speed is extremely important. Therefore, an appropriate accelerator is required to fulfill the requirements of these limitations. On the one hand, GPUs are widely used to accelerate the CNN under high power dissipation. On the other hand, the trend for FPGA implementation is increasing rapidly due to its low power consumption and facile re-configurability. In this work, we evaluate the inference performance of 10 classification models and 9 object detection models using the OpenVINO toolkit. In addition, we analyzed the implementation of these models on the DE5a-Net DDR4 equipped with an Arria 10 GX FPGA. The results show that the performance of Full-Precision FP32 classification models on a heterogeneous architecture FPGA/CPU is on average 3.6X faster than the CPU.**

***Keywords—deep-learning, heterogeneous architecture, FPGA, OpenVINO.***

## I. INTRODUCTION

Deep Learning techniques have developed rapidly in recent years and have been increasingly applied in a wide range of applications, including robot technology, computer vision, natural language processing, and autonomous vehicles. In which, CNN is actually one of the most used and popular DL networks, and is now one of the strongest image classification models [1]. Where many different CNN models are currently available for general applications, starting with the Alex-Net model [2] the first effective application of CNN, and later, with the achievement of AlexNet, many works have been published to enhance its performance. Some of them are VGGNet [3], GoogLeNet [4], Fast R-CNN [5], ResNet [6]. However, these algorithms are cumbersome and require weeks or more for the training phase. Thus, deploying a trained CNN model (inference) to perform predictions based on novel data becomes attractive to many deep learning researchers. To this end, Intel recently released an open-source CNN model inference engine named OpenVINO Toolkit [7].

This toolkit optimizes CNN models in the computer vision domain and promises to reach higher performance compared to deep learning frameworks used alone. To meet the above requirements (e.g., latency, memory...), it is possible to use accelerators, such as GPUs and FPGAs. On the one hand, GPUs have some limitations such as high power consumption, which restricts their use and adequacy in real-time and real-world applications, including, drones, robots, and autonomous cars [8]. On the other hand, field-programmable gate arrays (FPGAs) offer performance per watt equivalent to or greater than that achieved by modern GPUs [9]. In addition, FPGAs are characterized by their scalability and configurability, which makes them suitable for use in many different scenarios to solve problems posed by the limitations of GPUs without affecting the precision of the algorithm.

The main contribution of our paper is to evaluate the performance of the implementation of different deep learning inference models on a heterogeneous CPU / FPGA DE5a-Net DDR4 architecture using the OpenVINO toolkit.

The remaining sections of the paper are organized as follows: Section 2 surveys some related work. Section 3 introduces the OpenVINO toolkit. Section 4 presents the testbed architecture. Following that, Section 5 describes the evaluation of the implementation of deep learning inference models on CPU and FPGA. Finally, we conclude the paper in section 6.

## II. RELATED WORK

Many types of research have been carried out on the implementation of deep learning inference models on different accelerators. In [10], the authors evaluate the inference performance of 14 classification models with OpenVINO. The results show that the quantized models are faster on average 1.02X compared to the FP32 models on CPU, and the FP32 models on iGPU are 1.5X faster than the CPU. Furthermore, the work [11] focuses on using an Intel programmable accelerator board (PAC) with Arria 10 GX FPGA to accelerate the inference phase of DNN models, particularly HEP-CNN, CosmoGAN, and 3D-GAN. The authors use Intel OpenVINO and Intel Deep Learning (DLA) accelerator to accelerate the DNN models and optimize FPGA

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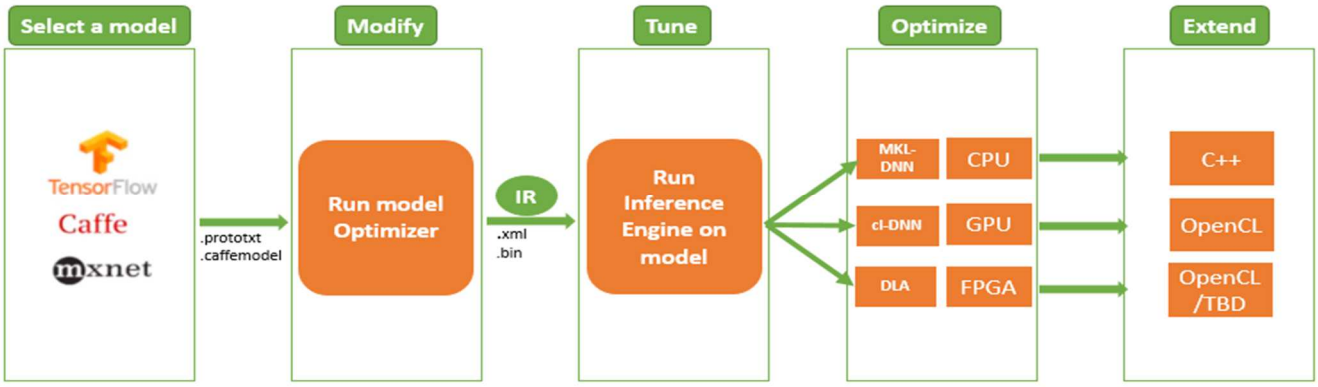



Fig. 1. OpenVINO workflow [14]

primitives. As a result, the speedup achieves 2.46X to 9.59X acceleration over a single core thread of a server class Skylake processor. In addition, the authors of the paper [12] propose a faster and more efficient embedded system for license plate segmentation and recognition. The authors used a modified single-shot detector (SSD) with a feature extractor that relies on depth separable convolutions and linear bottlenecks. The results on an Intel Xeon processor with 12 cores with 2.60 GHz per core reach 59ms of processing time while using OpenVINO the system runs on 14ms, and the processing time reaches 66ms on a Raspberry Pi3 and Intel Neural Computing stick 2 using OpenVINO. Also, the authors of the work [13] present a unified end-to-end solution for performing CNN model inference on embedded GPUs. The authors also perform the performance of three classification models using OpenVINO. The performance in terms of execution time (ms) of ResNet50_v1, MobileNet1.0, and SqueezeNet1.0 are 203.60, 53.48, 42.01 respectively. The above works show the success of deploying and accelerating deep learning inference models on different accelerator by using the OpenVINO toolkit.

## III. OpenVINO

OpenVINO for Open Visual Inference and Neural network Optimization is a toolkit designed by Intel to facilitate and accelerate the development of deep learning models from different frameworks (e.g. Caffe, Tensorflow...). OpenVINO handles models as a graph and employs high performance libraries such as Intel MKL-DNN for implementing graph operations and for improving the performance of CNN inference. In addition, this toolkit allows deep learning on hardware accelerators and heterogeneous platforms (CPU, GPU, FPGA) designed by Intel.

OpenVINO is composed, as shown in Figure 1, of two main parts: firstly, the intermediate representation (IR) is generated via Model Optimizer, which is a cross-platform to convert the pretrained model into OpenVINO format (.bin) and (.xml). The .bin file is a binary file that contains all the weights and biases, while the .xml file describes the topology, layers, connectivity, and parameters of the model. Secondly, the inference engine consists of a C++ library that provides a general API for reading the intermediate representation, defining the input and output formats, and executing the model on the devices [14].

In order to interact with the FPGA, the Deep Learning Accelerator (DLA) provides an API (DLIA plugin) that enables the inference of deep learning models on the FPGA through the interaction of the OpenVINO inference engine and the FPGA bitstream. DLA infers deep learning models as sub-graphs, consisting generally of a convolutional layer, max pooling, activation layer, and normalization layer.

Moreover, OpenVINO provides a heterogeneous plugin to compute inference models on multiple devices (FPGA, CPU), (GPU, CPU), etc. In our case, we evaluate the models on (FPGA, CPU) as described in Section 5.

## IV. Setup

For the purpose of evaluating the performance of using heterogeneous computing to implement the inference deep learning models, we chose the DE5a-Net DDR4, which enables faster data processing by a powerful Intel Arria 10 FPGA as shown in Figure 2. The board contains 16 GB DDR4 memory, running at more than 75 Gbps, and via PCIe Gen 3x8 between the board and the host PC, data transfer up to 7.876 Gbps [15]. In addition, the DE5a-Net DDR4 supports the OpenVINO toolkit for optimal computer vision and deep learning applications [16]. Further to these offerings, the host PC has an Intel® Xeon® Silver 4108 processor with eight cores running at a base frequency of 1.8 GHz. The host has 32GB of DDR4-2666 ECC SDRAM memory (2 x 16GB).

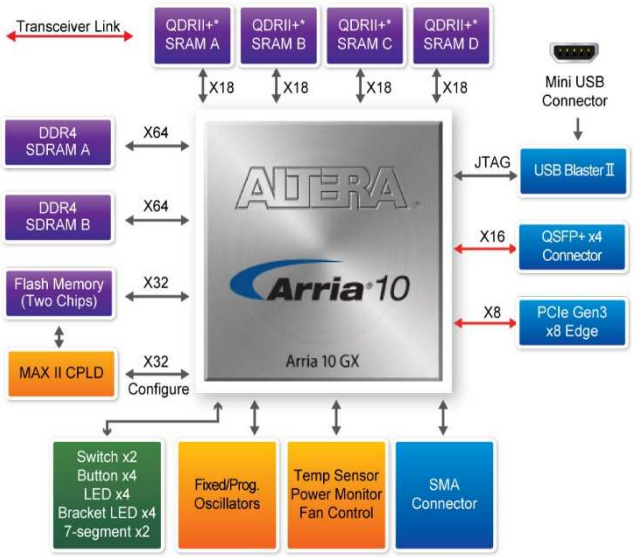



Fig. 2. DE5a-Net DDR4 architecture [15]

## V. Result and Discussion

In this paper, we selected 10 pre-trained Caffe models by open model Zoo for image classifications and 9 pre-trained

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Caffe models by Intel pre-trained model for object detection [17]. The evaluation of these inference deep learning models is performed on the DE5a-Net DDR4 board using OpenVINO toolkits. The following figures show the simulation results of different deep learning models on DE5a-Net DDR4.

Fig.3 shows the result of running the Person-detection-retail-0013 object detection model on FPGA/CPU. A video is chosen as input, after validation, the model detected people with desirable accuracy.




Fig. 3. Person-detection-retail-0013 model

Fig.4 shows the result of running Person-vehicle-bike-detection-crossroad-0078 object detection model in FPGA/CPU heterogeneous mode. We choose as input a video, after validation, the model detects both the vehicle and the person simultaneously.




Fig. 4. Person-vehicle-bike-detection-crossroad-0078 model

Moreover, other models operated successfully on DE5a-Net DDR4 as SqueezeNet v1.1 and VGG19, which are designed for image classification.

After the functional validation of these algorithms, we evaluate their performance by running the benchmark utility and comparing the performance on CPU and FPGA/CPU to find out the computational efficiency of FPGA for deep learning model acceleration.

The compute-intensive parts of the network execute on the Arria 10 FPGA accelerator, while the unsupported layers execute on the fallback CPU. Following the validation of the inference results of the classification and object detection models, we then execute the benchmark utility to perform latency and throughput measurements of the inference models.

Table I and Table II present the performance in terms of inference time and frames per second of the selected classification models on (CPU) and (FPGA/CPU) respectively using the OpenVINO toolkit.

Compared to FP32, the speedup on FPGA/CPU is on average 3.6X times faster than on CPU. Moreover, FP16 on FPGA/CPU is on average 0.9X times faster than FP32, the CPU does not support FP16.

TABLE I. THE INFERENCE TIME OF THE CLASSIFICATION MODELS

| | **CPU** | **FPGA/CPU** | |
|---|---|---|---|
| | ***FP32*** | ***FP32*** | ***FP16*** |
| **AlexNet** | **28.53** | **26.55** | **26.59** |
| **DenseNet 161** | **197.75** | **70.54** | **70.53** |
| **MobileNet v2** | **15.63** | **12.49** | **12.48** |
| **MobileNet v1 1.0 224** | **17.03** | **7.38** | **7.37** |
| SE-ResNet 50 | 135.64 | 206.03 | 204.50 |
| SE-ResNetXt 101 | 299.66 | 416.34 | 415.99 |
| **Inception (GoogleNet) v2** | **51.75** | **6.69** | **6.70** |
| **SqueezeNet v1.0** | **22.68** | **3.58** | **3.26** |
| **SqueezeNet v1.1** | **11.04** | **2.13** | **2.20** |
| **VGG19** | **469.66** | **71.65** | **71.66** |
| **Time unit** | **ms** | **ms** | **ms** |

TABLE II. THE FRAME PER SECOND OF THE CLASSIFICATION MODELS

| | **CPU** | **FPGA/CPU** | |
|---|---|---|---|
| | ***FP32*** | ***FP32*** | ***FP16*** |
| **AlexNet** | **69.85** | **75.26** | **74.90** |
| **DenseNet 161** | **10.08** | **28.30** | **28.31** |
| **MobileNet v2** | **123.57** | **159.05** | **159.14** |
| **MobileNet v1 1.0 224** | **114.49** | **269.12** | **268.96** |
| SE-ResNet 50 | 14.71 | 9.63 | 9.70 |
| SE-ResNetXt 101 | 6.65 | 4.79 | 4.79 |
| **Inception (GoogleNet) v2** | **37.97** | **294.10** | **294.13** |
| **SqueezeNet v1.0** | **86.70** | **672.23** | **668.51** |
| **SqueezeNet v1.1** | **170.68** | **895.56** | **865.60** |
| **VGG19** | **4.26** | **27.87** | **25.47** |

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In addition, Table III and IV show the inference time and frames per second performance of the selected object detection models on (CPU) and (FPGA/CPU) respectively using the OpenVINO toolkit.

Compared to FP32, the speedup on FPGA/CPU is on average 0.8 times faster than on CPU. In addition, FP16 on FPGA/CPU is on average 1.02 times faster than FP32, the CPU does not support FP16.

TABLE III. THE INFERENCE TIME OF THE OBJECT DETECTION MODELS

| | **CPU** | **FPGA/CPU** | |
|---|---|---|---|
| | ***FP32*** | ***FP32*** | ***FP16*** |
| **Face-detection-retail-0004** | **15.64** | **11.27** | **9.66** |
| **Face-detection-adas-0001** | **35.71** | **35.34** | **29.33** |
| **Pedestrian-and-vehicle-detector-adas-0001** | **42.67** | **35.71** | **35.06** |
| **Pedestrian-detection-adas-0002** | **35.54** | **34.57** | **33.86** |
| Person-detection-retail-0013 | 29.31 | 61.86 | 60.67 |
| **Vehicle-detection-adas-0002** | **34.87** | **31.74** | **33.90** |
| Vehicle-license-plate-detection-barrier-0106 | 8.53 | 16.98 | 18.31 |
| Person-vehicle-bike-detection-crossroad-0078 | 70.76 | 961.14 | 960.73 |
| Face-person-detection-retail-0002 | 38.28 | 71.18 | 70.85 |
| **Time unit** | **ms** | **ms** | **ms** |

TABLE IV. THE FRAME PER SECOND OF THE OBJECT DETECTION MODELS

| | **CPU** | **FPGA/CPU** | |
|---|---|---|---|
| | ***FP32*** | ***FP32*** | ***FP16*** |
| **Face-detection-retail-0004** | **63.93** | **88.69** | **103.46** |
| **Face-detection-adas-0001** | **27.99** | **28.28** | **34.09** |
| **Pedestrian-and-vehicle-detector-adas-0001** | **23.43** | **28.00** | **28.52** |
| **Pedestrian-detection-adas-0002** | **28.13** | **28.92** | **29.53** |
| Person-detection-retail-0013 | 34.11 | 16.16 | 16.48 |
| **Vehicle-detection-adas-0002** | **28.67** | **31.49** | **29.49** |
| Vehicle-license-plate-detection-barrier-0106 | 117.12 | 58.88 | 54.59 |
| Person-vehicle-bike-detection-crossroad-0078 | 14.13 | 1.04 | 1.04 |
| Face-person-detection-retail-0002 | 26.11 | 14.04 | 14.11 |

From the results in Table I, II, III, and IV, we notice that some models are accelerated at the FPGA level such as classification models: AlexNet, DenseNet 161, MobileNet v2, MobileNet v1 1.0 224, Inception (GoogleNet) v2, SqueezeNet v1.0, SqueezeNet v1.1, and VGG19, also object detection models including, face-detection-retail-0004, Face-detection-adas-0001,Pedestrian-and-vehicle-detector-adas-0001, Pedestrian-detection-adas-0002, and Vehicle-detection-adas-0002.

While other models remain fast at the CPU level like SE-ResNet 50, SE-ResNetXt 101 for classification models, and person-detection-retail-0013, Vehicle-license-plate-detection-barrier-0106, Person-vehicle-bike-detection-crossroad-0078, Face-person-detection-retail-0002 for object detection models.

In heterogeneous mode, the heavy part runs on the FPGA and the other part on the CPU, therefore in case the networks contain activation functions that cannot be supported by the FPGA, the data transmission between the FPGA and the CPU in heterogeneous mode can take more time than the normal mode. Moreover, some blocks in the networks have a sequential nature that requires rewriting for parallel execution on FPGA.

In addition, Table III and IV show that there is no significant acceleration on FPGA / CPU for object detection models compared to image classification models, the evaluation gives satisfactory results at the CPU level.

## VI. CONCLUSION

The OpenVINO toolkit enables the development of cross-platform solutions to computer vision problems based on image processing, machine learning, and deep learning. This toolkit offers a broad range of algorithms optimized for optimal performance on Intel hardware (Intel processors, Intel GPUs, Intel Movidius Neural Compute Stick, and Intel FPGA). Moreover, it is possible to run algorithms heterogeneously on various Intel accelerators using the same application-programming interface.

In our work, we focus on the performance evaluation of inference time and throughput for different deep learning models using OpenVINO. As a result, OpenVINO provides better performance in accelerating deep learning algorithms, where implementing heterogeneous FPGA/CPU architectures provides on average 3.5 times faster acceleration than a homogeneous CPU architecture for classification models. However, object detection models achieve better results on the CPU with the OpenVINO toolkit.

For our future work, we aim to optimize such object detection models using high-level synthesis techniques.

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